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Semiconductor structure with wrapped-around backside contact

Abstract

A semiconductor structure includes a backside contact, and an unmerged source/drain region. The backside contact is wrapped-around the unmerged source/drain region.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9570395	12/2016	Sengupta et al.	N/A	N/A
9601586	12/2016	Kittl et al.	N/A	N/A
10586765	12/2019	Smith et al.	N/A	N/A
10636739	12/2019	Beyne et al.	N/A	N/A
10700207	12/2019	Chen et al.	N/A	N/A
10797139	12/2019	Morrow et al.	N/A	N/A
11222892	12/2021	Su et al.	N/A	N/A
2017/0194143	12/2016	Balakrishnan	N/A	H01L 21/02603
2018/0108526	12/2017	Mitard	N/A	H10D 30/43
2019/0057867	12/2018	Smith	N/A	H01L 21/02603
2019/0305106	12/2018	Bi	N/A	H10D 30/6212
2020/0105671	12/2019	Lai et al.	N/A	N/A
2021/0111115	12/2020	Morrow et al.	N/A	N/A
2021/0134721	12/2020	Chiang et al.	N/A	N/A
2021/0193807	12/2020	Guha	N/A	H10D 64/01
2021/0202385	12/2020	Huang et al.	N/A	N/A
2021/0305252	12/2020	Chiang et al.	N/A	N/A
2021/0305381	12/2020	Chiang et al.	N/A	N/A
2021/0351303	12/2020	Ju et al.	N/A	N/A
2021/0376071	12/2020	Liu et al.	N/A	N/A
2021/0376093	12/2020	Chu et al.	N/A	N/A
2021/0399099	12/2020	Chu et al.	N/A	N/A
2022/0069117	12/2021	Yu et al.	N/A	N/A
2022/0165848	12/2021	Lin et al.	N/A	N/A
2022/0165856	12/2021	Yu et al.	N/A	N/A
2022/0344472	12/2021	Lin	N/A	H10D 30/6211

OTHER PUBLICATIONS

A. Gupta et al., “Buried Power Rail Integration with Si FinFETs for CMOS Scaling beyond the 5 nm Node,” 2020 IEEE Symposium on VLSI Technology, Jun. 16-19, 2020, 2 pages. cited by applicant

D. Prasad et al., “Buried Power Rails and Back-side Power Grids: Arm CPU Power Delivery Network Design Beyond 5nm,” 2019 IEEE International Electron Devices Meeting (IEDM), Dec. 7-11, 2019, pp. 19.1.1-19.1.4. cited by applicant

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Background/Summary

BACKGROUND

(1) A field-effect transistor (FET) is a transistor having a source, a gate, and a drain, and having action that depends on the flow of carriers (electrons or holes) along a channel that runs between the source and drain. Current through the channel between the source and drain may be controlled by a transverse electric field under the gate.

(2) FETs are widely used for switching, amplification, filtering, and other tasks. FETs include metal-oxide-semiconductor FETs (MOSFETs). Complementary MOS (CMOS) devices are widely used, where both n-type and p-type transistors (nFET and pFET) are used to form logic and other circuitry. Source and drain regions of a FET are typically formed by adding dopants to target regions of a semiconductor body on either side of a channel, with the gate being formed above the channel. The gate includes a gate dielectric over the channel and a gate conductor over the gate dielectric. The gate dielectric is an insulator material that prevents large leakage current from flowing into the channel when voltage is applied to the gate conductor while allowing applied gate voltage to produce a transverse electric field in the channel.

SUMMARY

(3) Illustrative embodiments of the present application include techniques for use in semiconductor manufacture. In an illustrative embodiment, a semiconductor structure comprises a backside contact, and an unmerged source/drain region. The backside contact is wrapped-around the unmerged source/drain region.

(4) In another illustrative embodiment, a semiconductor structure comprises a backside contact, a first unmerged source/drain region, a frontside contact, and a second unmerged source/drain region. The backside contact is wrapped-around the first unmerged source/drain region. The frontside contact is wrapped-around the second unmerged source/drain region.

(5) In yet another illustrative embodiment, an integrated circuit comprises one or more semiconductor structures. At least one of the one or more semiconductor structures comprises a backside contact, a first unmerged source/drain region, a frontside contact; and a second unmerged source/drain region. The backside contact is wrapped-around the first unmerged source/drain region. The frontside contact is wrapped-around the second unmerged source/drain region.

(6) Other embodiments will be described in the following detailed description of embodiments, which is to be read in conjunction with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a cross sectional view illustrating a semiconductor structure for use at a first-intermediate fabrication stage, according to an illustrative embodiment.

(2) FIG. 2A depicts a top view of a semiconductor structure for use at a second-intermediate fabrication stage, according to an illustrative embodiment.

(3) FIG. 2B depicts a side cross-sectional view of the semiconductor structure taken along the X-X axis of FIG. 2A for use at the second-intermediate fabrication stage, according to an illustrative embodiment.

(4) FIG. 2C depicts a side cross-sectional view of the semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the second-intermediate fabrication stage, according to an illustrative embodiment.

(5) FIG. 3A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a third-intermediate fabrication stage, according to an illustrative embodiment.

(6) FIG. 3B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y

axis of FIG. 2A for use at the third-intermediate fabrication stage, according to an illustrative embodiment.

(7) FIG. 4A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a fourth-intermediate fabrication stage, according to an illustrative embodiment.

(8) FIG. 4B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the fourth-intermediate fabrication stage, according to an illustrative embodiment.

(9) FIG. 5A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a fifth-intermediate fabrication stage, according to an illustrative embodiment.

(10) FIG. 5B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the fifth-intermediate fabrication stage, according to an illustrative embodiment.

(11) FIG. 6A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a sixth-intermediate fabrication stage, according to an illustrative embodiment.

(12) FIG. 6B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the sixth-intermediate fabrication stage, according to an illustrative embodiment.

(13) FIG. 7A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a seventh-intermediate fabrication stage, according to an illustrative embodiment.

(14) FIG. 7B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the seventh-intermediate fabrication stage, according to an illustrative embodiment.

(15) FIG. 8A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at an eighth-intermediate fabrication stage, according to an illustrative embodiment.

(16) FIG. 8B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the eighth-intermediate fabrication stage, according to an illustrative embodiment.

(17) FIG. 9A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a ninth-intermediate fabrication stage, according to an illustrative embodiment.

(18) FIG. 9B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the ninth-intermediate fabrication stage, according to an illustrative embodiment.

(19) FIG. 10A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a tenth-intermediate fabrication stage, according to an illustrative embodiment.

(20) FIG. 10B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the tenth-intermediate fabrication stage, according to an illustrative embodiment.

(21) FIG. 11A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at an eleventh-intermediate fabrication stage, according to an illustrative embodiment.

(22) FIG. 11B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the eleventh-intermediate fabrication stage, according to an illustrative embodiment.

- (23) FIG. 12A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a twelfth-intermediate fabrication stage, according to an illustrative embodiment.
- (24) FIG. 12B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the twelfth-intermediate fabrication stage, according to an illustrative embodiment.
- (25) FIG. 13A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a thirteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (26) FIG. 13B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the thirteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (27) FIG. 14A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a fourteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (28) FIG. 14B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the fourteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (29) FIG. 15A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a fifteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (30) FIG. 15B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the fifteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (31) FIG. 16A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a sixteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (32) FIG. 16B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the sixteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (33) FIG. 17A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a seventeenth-intermediate fabrication stage, according to an illustrative embodiment.
- (34) FIG. 17B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the seventeenth-intermediate fabrication stage, according to an illustrative embodiment.
- (35) FIG. 18A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at an eighteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (36) FIG. 18B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the eighteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (37) FIG. 19A depicts a side cross-sectional view of a semiconductor structure taken along the X-X axis of FIG. 2A for use at a nineteenth-intermediate fabrication stage, according to an illustrative embodiment.
- (38) FIG. 19B depicts a side cross-sectional view of a semiconductor structure taken along the Y-Y axis of FIG. 2A for use at the nineteenth-intermediate fabrication stage, according to an illustrative embodiment.

DETAILED DESCRIPTION

- (39) Illustrative embodiments of the invention may be described herein in the context of illustrative

methods for forming a backside contact that is wrapped-around an unmerged source/drain region in semiconductor structures to prevent a higher resistance-capacitance product (RC), along with illustrative apparatus, systems and devices formed using such methods. However, it is to be understood that embodiments of the invention are not limited to the illustrative methods, apparatus, systems and devices but instead are more broadly applicable to other suitable methods, apparatus, systems and devices. The term “unmerged” illustratively refers to source/drain regions that are grown on an end of sidewalls of nanosheet channel layers in an opening that do not connect with the source/drain regions grown on the end of opposing sidewalls of nanosheet channel layers in the opening.

(40) It is to be understood that the various features shown in the accompanying drawings are schematic illustrations that are not necessarily drawn to scale. Moreover, the same or similar reference numbers are used throughout the drawings to denote the same or similar features, elements, or structures, and thus, a detailed explanation of the same or similar features, elements, or structures will not be repeated for each of the drawings. Further, the terms “exemplary” and “illustrative” as used herein mean “serving as an example, instance, or illustration.” Any embodiment or design described herein as “exemplary” or “illustrative” is not to be construed as preferred or advantageous over other embodiments or designs.

(41) Furthermore, it is to be understood that the embodiments discussed herein are not limited to the particular materials, features, and processing steps shown and described herein. In particular, with respect to semiconductor processing steps, it is to be emphasized that the descriptions provided herein are not intended to encompass all of the processing steps that may be required to form a functional semiconductor integrated circuit device. Rather, certain processing steps that are commonly used in forming semiconductor devices, such as, for example, wet cleaning and annealing steps, are purposefully not described herein for economy of description. It is to be understood that the terms “about” or “substantially” as used herein with regard to thicknesses, widths, percentages, ranges, etc., are meant to denote being close or approximate to, but not exactly. For example, the term “about” or “substantially” as used herein implies that a small margin of error may be present, such as 1% or less than the stated amount.

(42) Reference in the specification to “one embodiment” or “an embodiment” of the present principles, as well as other variations thereof, means that a particular feature, structure, characteristic, and so forth described in connection with the embodiment is included in at least one embodiment of the present principles. Thus, the appearances of the phrase “in one embodiment” or “in an embodiment”, as well as any other variations, appearing in various places throughout the specification are not necessarily all referring to the same embodiment. The term “positioned on” means that a first element, such as a first structure, is present on a second element, such as a second structure, wherein intervening elements, such as an interface structure, e.g., interface layer, may be present between the first element and the second element. The term “direct contact” means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary conducting, insulating or semiconductor layers at the interface of the two elements.

(43) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the scope of the present concept.

(44) As used herein, “height” refers to a vertical size of an element (e.g., a layer, trench, hole, opening, etc.) in the cross-sectional views measured from a bottom surface to a top surface of the element, and/or measured with respect to a surface on which the element is located. Conversely, a “depth” refers to a vertical size of an element (e.g., a layer, trench, hole, opening, etc.) in the cross-sectional views measured from a top surface to a bottom surface of the element. Terms such as “thick”, “thickness”, “thin” or derivatives thereof may be used in place of “height” where

indicated.

(45) As used herein, “width” or “length” refers to a size of an element (e.g., a layer, trench, hole, opening, etc.) in the drawings measured from a side surface to an opposite surface of the element. Terms such as “thick”, “thickness”, “thin” or derivatives thereof may be used in place of “width” or “length” where indicated.

(46) In the IC chip fabrication industry, there are three sections referred to in a typical IC chip build: front-end-of-line (FEOL), back-end-of-line (BEOL), and the section that connects those two together, the middle-of-line (MOL). The FEOL is made up of the semiconductor devices, e.g., transistors, the BEOL is made up of interconnects and wiring, and the MOL is an interconnect between the FEOL and BEOL that includes material to prevent the diffusion of BEOL metals to FEOL devices. Accordingly, illustrative embodiments described herein may be directed to BEOL semiconductor processing and structures. BEOL is the second portion of IC fabrication where the individual devices (e.g., transistors, capacitors, resistors, etc.) become interconnected with wiring on the wafer, e.g., the metallization layer or layers. BEOL includes contacts, insulating layers (dielectrics), metal levels, and bonding sites for chip-to-package connections. In the BEOL, part of the fabrication stage contacts (pads), interconnect wires, vias and dielectric structures are formed. For modern IC processes, more than 10 metal layers may be added in the BEOL.

(47) Embodiments described below may be applicable to FEOL processing and structures, BEOL processing and structures, or both FEOL and BEOL processing and structures. In particular, although an exemplary processing scheme may be illustrated using a FEOL processing scenario, such approaches may also be applicable to BEOL processing. Likewise, although an exemplary processing scheme may be illustrated using a BEOL processing scenario, such approaches may also be applicable to FEOL processing.

(48) One concern in forming a backside contact is a higher resistance-capacitance product due to present semiconductor processing which cannot perform a high temperature anneal when a BEOL interconnect is formed in a frontside of a semiconductor substrate. Illustrative embodiments provide methods and structures for overcoming the foregoing drawbacks by forming a wrapped-around backside contact. Referring now to FIGS. 1-19B, FIG. 1 shows a semiconductor structure **100** having substrate **102** and etch stop layer **104** formed in the substrate **102**. The substrate **102** may be formed of any suitable semiconductor structure, including various silicon-containing materials including but not limited to silicon (Si), silicon germanium (SiGe), silicon germanium carbide (SiGeC), silicon carbide (SiC) and multi-layers thereof. Although silicon is the predominantly used semiconductor material in wafer fabrication, alternative semiconductor materials can be employed as additional layers, such as, but not limited to, germanium (Ge), gallium arsenide (GaAs), gallium nitride (GaN), SiGe, cadmium telluride (CdTe), zinc selenide (ZnSe), etc. In one illustrative embodiment, substrate **102** is silicon.

(49) The etch stop layer **104** may comprise a buried oxide (BOX) layer or silicon germanium (SiGe), or another suitable material such as a III-V semiconductor epitaxial layer.

(50) Nanosheets are formed over the substrate **102**, where the nanosheets include sacrificial layer **106**, sacrificial layers **108-1**, **108-2** and **108-3** (collectively, sacrificial layers **108**) and nanosheet channel layers **110-1**, **110-2** and **110-3** (collectively, nanosheet channel layers **110**).

(51) The sacrificial layer **106** and sacrificial layers **108** are illustratively formed of different sacrificial materials, such that they may be etched or otherwise removed selective to one another. In some embodiments, both the sacrificial layer **106** and sacrificial layers **108** are formed of SiGe, but with different percentages of Ge. For example, the sacrificial layer **106** may have a relatively higher percentage of Ge (e.g., 55% Ge), and the sacrificial layers **108** may have a relatively lower percentage of Ge (e.g., 25% Ge). Other combinations of different sacrificial materials may be used in other embodiments.

(52) The nanosheet channel layers **110** may be formed of Si or another suitable material (e.g., a material similar to that used for the substrate **102**).

(53) Although three layers each of the sacrificial layers **108** and the nanosheet channel layers **110** are shown, the number of sacrificial layers **108** and the nanosheet channel layers **110** should not be considered limiting and any number are contemplated.

(54) FIGS. 2A-2C show semiconductor structure **100** at a second-intermediate fabrication stage. During this stage, nanosheet patterning and formation of FET stacks **109a**, **109b**, **109c** and **109d** together with shallow trench isolation (STI) regions **112** and dummy gates **114** are carried out. Each of FET stacks **109a**, **109b**, **109c** and **109d** contain a respective FET device. However, this is merely illustrative and it is contemplated that FET stacks **109a**, **109b**, **109c** and **109d** can contain any number of FET devices. The FET stacks **109a**, **109b**, **109c** and **109d** may comprise nFET devices or pFET devices or combinations thereof.

(55) FIG. 2B shows semiconductor structure **100** having dummy gate **114** and hardmask layer **116** on the top most nanosheet channel layer **110-3**. Suitable dummy gate material includes, for example, polycrystalline silicon, amorphous silicon or microcrystal silicon. Hardmask layer **116** can be composed of a flowable organic material such as, for example, a spin-on-carbon (SOC), Si.sub.3N.sub.4, SiBCN, SiNC, SiN, SiCO, SiO.sub.2, and SiNOC. The dummy gate can be formed by conventional lithographic and etching processes of hardmask layer **116**.

(56) FIG. 2C shows semiconductor structure **100** having STI regions **112**. The STI regions **112** may be formed by patterning a hardmask layer (not shown), followed by etching exposed portions of the nanosheet channel layers **110**, the sacrificial layers **108** and **106**, and through a portion of the substrate **102**. A dielectric material is then deposited to form STI regions **112**, followed by a planarization step such as a chemical mechanical planarization (CMP) process and recess. The STI regions **112** may be formed of a suitable dielectric material such as silicon dioxide (SiO.sub.2), silicon oxycarbide (SiOC), silicon oxynitride (SiON), etc. Following formation of STI regions **112**, the patterned hardmask layer (not shown) is also removed.

(57) FIGS. 3A and 3B show semiconductor structure **100** at a third-intermediate fabrication stage. During this stage, bottom dielectric isolation (BDI) layer **118** and sidewall spacers **120** are formed. BDI layer **118** can be formed by selectively removing sacrificial layer **106** without etching the sacrificial layers **108** and nanosheet channel layers **110** in the stack. That is, the sacrificial layer **106** can be etched/removed without removing the sacrificial layers **108** alternatingly formed with nanosheet channel layers **110**. After removal of sacrificial layer **106**, the cavity is replaced with a dielectric fill to form BDI layer **118**. In one embodiment, the BDI layer **118** can be formed during deposition of sidewall spacers **120**. In some other embodiments, the BDI layer **118** can be a different material than sidewall spacers **120**. In one illustrative embodiment, the BDI layer **118** and sidewall spacers **120** can independently be a nitride or an oxynitride such as, for example, Si.sub.3N.sub.4, SiBCN, SiNC, SiN, SiCO, SiO.sub.2 and SiNOC. In some exemplary embodiments, sidewall spacers **120** can include a material that is resistant to some etching processes such as, for example, HF chemical etching or chemical oxide removal etching. The BDI layer **118** and sidewall spacers **120** can be formed using conventional deposition techniques such as by physical vapor deposition (PVD), atomic layer deposition (ALD), chemical vapor deposition (CVD), etc.

(58) FIGS. 4A and 4B show semiconductor structure **100** at a fourth-intermediate fabrication stage. During this stage, interlayer dielectric (ILD) layer **122** is formed on STI regions **112** and between adjacent sidewall spacers **120**. ILD layer **122** can be formed of any suitable dielectric material, such as SiO.sub.2, SiOC, SiON, etc. The ILD layer **122** can be formed using conventional deposition techniques such as by PVD, ALD, CVD, etc. Next, an isotropic etch back is carried out on ILD layer **122** so that the top surface of ILD layer **122** is below the top surface of sidewall spacers **120**.

(59) FIGS. 5A and 5B show semiconductor structure **100** at a fifth-intermediate fabrication stage. During this stage, inner spacers **124** and openings **126** are formed. Openings **126** are formed by selectively etching a portion of sacrificial layers **108** and nanosheet channel layers **110** until a top

surface of BDI layer **118** is reached. Suitable etching processes include, for example, reactive ion etching (ME). The inner spacers **124** may be formed by filling indent spaces resulting from selective indent etches of the sacrificial layers **108**. The inner spacers **124** may be formed of silicon nitride (SiN) or another suitable material such as SiBCN, silicon carbide oxide (SiCO), SiOCN, etc. The inner spacers **124** can be formed using conventional deposition techniques such as by PVD, ALD, CVD, etc., followed by an isotropic etching back of the dielectric liner. The inner spacers **124** may have heights matching that of the sacrificial layers **108**.

(60) FIGS. **6A** and **6B** show semiconductor structure **100** at a fifth-intermediate fabrication stage. During this stage, one of openings **126** is extended through substrate **102** and exposing a top surface of etch stop layer **104** to provide backside contact openings **126'**. Backside contact openings **126'** are formed by first depositing a mask layer **128** (such as an organic planarization layer (OPL) or a spin-on-carbon (SOC)) on semiconductor structure **100** using any conventional deposition process such spin-on coating or any other suitable deposition process. Next, the mask layer **128** is patterned and then selectively etched to form backside contact openings **126'**.

(61) FIGS. **7A** and **7B** show semiconductor structure **100** at a sixth-intermediate fabrication stage. During this stage, a sacrificial placeholder layer **130** is formed in backside contact openings **126'** and below a top surface of BDI layer **118** using any conventional deposition process such as PVD, ALD and CVD, or bottom-up epitaxial growth process. Sacrificial placeholder layer **130** may be formed of a sacrificial material or materials, such as titanium oxide (TiO.sub.x), aluminum oxide (AlO.sub.x), silicon carbide (SiC), SiGe, III-V materials, etc.

(62) FIGS. **8A** and **8B** show semiconductor structure **100** at an eighth-intermediate fabrication stage. During this stage, mask layer **128** is removed, for example, by an ash etching process, and openings **126** are formed again. Next, unmerged source/drain regions **132** are formed on nanosheet channel layers **110**. Unmerged source/drain regions **132** may be formed using epitaxial growth processes. In non-limiting illustrative embodiments, the source/drain regions can be grown on nanosheet channel layers **110** until the source/drain regions reach a predetermined size, where the source/drain regions remain separate unmerged source/drains. In illustrative embodiments, the unmerged source/drain regions **132** are grown from ends of the sidewalls of nanosheet channel layers **110** in each of openings **126** and **126'** and do not connect with each other. For example, the unmerged source/drain regions **132** grown on one end of sidewalls of nanosheet channel layers **110** in each of openings **126** and **126'** do not connect with the unmerged source/drain regions **132** grown on the other end of sidewalls of nanosheet channel layers **110** in each of openings **126** and **126'** between neighboring replacement gate structures **136** as discussed below.

(63) Unmerged source/drain regions **132** may be suitably doped, such as using ion implantation, gas phase doping, plasma doping, plasma immersion ion implantation, cluster doping, infusion doping, liquid phase doping, solid phase doping, etc. N-type dopants may be selected from a group of phosphorus (P), arsenic (As) and antimony (Sb), and p-type dopants may be selected from a group of boron (B), boron fluoride (BF.sub.2), gallium (Ga), indium (In), and thallium (Tl). In some embodiments, the epitaxy process comprises in-situ doping (dopants are incorporated in epitaxy material during epitaxy). Epitaxial materials may be grown from gaseous or liquid precursors.

(64) Epitaxial materials may be grown using vapor-phase epitaxy (VPE), molecular-beam epitaxy (MBE), liquid-phase epitaxy (LPE), rapid thermal chemical vapor deposition (RTCVD), metal organic chemical vapor deposition (MOCVD), ultra-high vacuum chemical vapor deposition (UHVCVD), low-pressure chemical vapor deposition (LPCVD), limited reaction processing CVD (LRPCVD), or other suitable processes. Epitaxial silicon, silicon germanium (SiGe), germanium (Ge), and/or carbon doped silicon (Si:C) silicon can be doped during deposition (in-situ doped) by adding dopants, such as n-type dopants (e.g., phosphorus or arsenic) or p-type dopants (e.g., boron or gallium), depending on the type of transistor to be formed. The dopant concentration in the source/drain can range from $1 \times 10^{19} \text{ cm}^{-3}$ to $3 \times 10^{21} \text{ cm}^{-3}$, or preferably

between 2×10.sup.20 cm.sup.-3 to 3×10.sup.20 cm.sup.-3.

(65) FIGS. **9A** and **9B** show semiconductor structure **100** at a ninth-intermediate fabrication stage. During this stage, additional sacrificial placeholder layer **130** is formed in openings **126** and **126'** using any conventional deposition process such as PVD, ALD and CVD. Sacrificial placeholder layer **130** can then be subjected to a planarization process such as CMP or other suitable planarization process. As depicted in FIG. **9A**, additional sacrificial placeholder layer **130** is selectively recessed to expose sidewall spacers **120** using any suitable wet or dry etch.

Alternatively, the sacrificial placeholder layer **130** can also be formed by selectively epi growth.

(66) FIGS. **10A** and **10B** show semiconductor structure **100** at a tenth-intermediate fabrication stage. During this stage, ILD layer **134** is formed on unmerged source/drain regions **132**, followed by a planarization process such as CMP. ILD layer **134** can be formed by a similar process and of a similar material as ILD layer **122**. Next, exposed sidewall spacers **120** and hardmask layer **116** are selectively removed using any suitable etching process.

(67) One or more wet or dry etching processes are then performed to remove the dummy gate **114** and sacrificial layers **108** to thereby define a gate cavity where a replacement gate structure will subsequently be formed for the semiconductor structure **100**. Replacement gate structures **136** are then formed in the gate cavity. The replacement gate structures **136** depicted herein are intended to be representative in nature of any type of gate structure that may be employed in manufacturing integrated circuit products using so-called gate-last (replacement gate) manufacturing techniques. The replacement gate structure typically comprises a high-k (e.g., a k value greater than 5 or greater than 10) gate insulation layer (not individually shown), such as hafnium oxide, one or more metal layers (not individually shown) (e.g., layers of titanium nitride or TiAlC depending upon the type of transistor device being manufactured), and a bulk conductive material layer (not individually shown), such as tungsten or aluminum.

(68) FIGS. **11A** and **11B** show semiconductor structure **100** at an eleventh-intermediate fabrication stage. During this stage, frontside middle-of-the-line contact openings **138** are formed by first depositing an additional amount of ILD layer **134** on semiconductor structure **100**, followed by a planarization process such as CMP. Next, ILD layer **134** is subjected to a conventional lithographic and etching process to form frontside middle-of-the-line contact openings **138**. The frontside middle-of-the-line contact openings **138** are formed to expose the sacrificial placeholder layer **130**. In addition, ILD layer **134** remains over sacrificial placeholder layer **130** formed in backside contact openings **126'** discussed above.

(69) FIGS. **12A** and **12B** show semiconductor structure **100** at a twelfth-intermediate fabrication stage. During this stage, sacrificial placeholder layer **130** present in frontside middle-of-the-line contact openings **138** is selectively removed using any suitable etch processing that removes the material of the sacrificial placeholder layer **130** selective to that of the rest of the structure. A suitable etching process include, for example, RIE.

(70) FIGS. **13A** and **13B** show semiconductor structure **100** at a thirteenth-intermediate fabrication stage following formation of a metallization layer and frontside back-end-of-line (BEOL) interconnects and following bonding of the structure to a carrier wafer. During this stage, frontside middle-of-the-line contacts **140** are formed by depositing a high conductance metal using any conventional technique such as ALD, CVD, PVD, and/or plating. Suitable contact metals include, a silicide liner such as Ti, Ni, or NiPt, etc, a thin adhesion metal liner, such as TiN, and high conductive metal fills, such as, for example, tungsten (W), aluminum (Al), copper (Cu), cobalt (Co), ruthenium (Ru), molybdenum (Mo), or any other suitable conductive material. The contact metals can be planarized using, for example, a planarizing process such as CMP. Other planarization processes can include grinding and polishing. As shown in FIG. **13B**, frontside middle-of-the-line contacts **140** are wrapped around unmerged source/drain regions **132**. Thus, frontside middle-of-the-line contacts **140** can also be referred to as wrapped-around frontside contacts.

(71) Frontside BEOL interconnect **142** includes various BEOL interconnect structures. For example, frontside BEOL interconnect **142** is a metallization structure that includes one or more metal layers disposed on a side of semiconductor structure **100** opposite of the side on which the backside BEOL metallization structure is disposed. The metal layers of the frontside BEOL interconnect **142** each have metal lines for making interconnections to the semiconductor device.

(72) Carrier wafer **144** may be formed of materials similar to that of the substrate **102**, and may be formed over the frontside BEOL interconnect **142** using a wafer bonding process, such as dielectric-to-dielectric bonding.

(73) FIGS. **14A** and **14B** show semiconductor structure **100** at a fourteenth-intermediate fabrication stage. During this stage, backside processing of the substrate **102** is carried out. This can be accomplished, for example, by flipping the structure over so that the backside of the substrate **102** (i.e., the back surface) is facing up. First, portions of the substrate **102** may be removed from the backside using, for example, a combination of wafer grinding, CMP, dry etch and wet etch to selectively remove substrate **102** until the etch stop layer **104** is reached.

(74) FIGS. **15A** and **15B** illustrate semiconductor structure **100** at a fifteenth-intermediate fabrication stage. During this stage, the etch stop layer **104** is selectively removed using, for example, a wet etch to selectively remove etch stop layer **104** until substrate **102** is reached. The remaining portions of the substrate **102** are removed utilizing a selective etch process such as a wet etch.

(75) FIGS. **16A** and **16B** illustrate semiconductor structure **100** at a sixteenth-intermediate fabrication stage. During this stage, backside ILD layer **146** may be formed of similar processes and material as ILD layer **122**. The material of the backside ILD layer **146** may initially be overfilled, followed by planarization (e.g., using CMP).

(76) FIGS. **17A** and **17B** illustrate semiconductor structure **100** at a seventeenth-intermediate fabrication stage. During this stage, sacrificial placeholder layer **130** is selectively removed to form a backside contact opening **148** using any suitable etch processing that removes the material of the sacrificial placeholder layer **130** selective to that of the rest of the structure. A suitable etching process include, for example, ME.

(77) FIGS. **18A** and **18B** illustrate semiconductor structure **100** at an eighteenth-intermediate fabrication stage. During this stage, backside contacts **150** are formed by depositing a suitable metal in the backside contact opening **148**. A suitable metal can be any of the metals discussed above for frontside middle-of-the-line contact **140**. As shown in FIG. **18B**, backside contacts **150** are wrapped around unmerged source/drain regions **132**. Thus, backside contacts **150** can also be referred to as wrapped-around backside contacts.

(78) FIGS. **19A** and **19B** illustrate semiconductor structure **100** at a nineteenth-intermediate fabrication stage. During this stage, backside BEOL interconnect **152** is formed on backside ILD layer **146** and backside contacts **150** using similar processes and similar structures as frontside BEOL interconnect **142**.

(79) Semiconductor devices and methods for forming the same in accordance with the above-described techniques can be employed in various applications, hardware, and/or electronic systems. Suitable hardware and systems for implementing embodiments of the invention may include, but are not limited to, personal computers, communication networks, electronic commerce systems, portable communications devices (e.g., cell and smart phones), solid-state media storage devices, functional circuitry, etc. Systems and hardware incorporating the semiconductor devices are contemplated embodiments of the invention. Given the teachings provided herein, one of ordinary skill in the art will be able to contemplate other implementations and applications of embodiments of the invention.

(80) In some embodiments, the above-described techniques are used in connection with semiconductor devices that may require or otherwise utilize, for example, CMOSs, MOSFETs, and/or FinFETs. By way of non-limiting example, the semiconductor devices can include, but are

not limited to CMOS, MOSFET, and FinFET devices, and/or semiconductor devices that use CMOS, MOSFET, and/or FinFET technology.

(81) Various structures described above may be implemented in integrated circuits. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher-level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either: (a) an intermediate product, such as a motherboard, or (b) an end product. The end product can be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

(82) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A semiconductor structure, comprising: a backside contact; a frontside interlevel dielectric layer disposed on a top surface of the backside contact; and an unmerged source/drain region; wherein the backside contact is wrapped-around the unmerged source/drain region.
2. The semiconductor structure according to claim 1, wherein a portion of the backside contact is disposed in a backside interlayer dielectric layer.
3. The semiconductor structure according to claim 2, wherein another portion of the backside contact is disposed on the unmerged source/drain region.
4. The semiconductor structure according to claim 1, wherein the unmerged source/drain region is disposed on opposing sidewalls of nanosheet channel layers.
5. The semiconductor structure according to claim 1, further comprising: a gate structure disposed adjacent the unmerged source/drain region.
6. The semiconductor structure according to claim 5, wherein the gate structure is part of a stack of nanosheet channel layers.
7. The semiconductor structure according to claim 6, wherein the unmerged source/drain region is disposed on opposing sidewalls of each of the nanosheet channel layers.
8. The semiconductor structure according to claim 2, further comprising: a backside interconnect structure disposed on the backside contact and the backside interlayer dielectric layer.
9. The semiconductor structure according to claim 5, further comprising a frontside interconnect structure disposed over the gate structure.
10. A semiconductor structure, comprising: a backside contact; a frontside interlevel dielectric layer disposed on a top surface of the backside contact; a first unmerged source/drain region; a frontside contact; and a second unmerged source/drain region; wherein the backside contact is wrapped-around the first unmerged source/drain region; and wherein the frontside contact is wrapped-around the second unmerged source/drain region.
11. The semiconductor structure according to claim 10, wherein a portion of the backside contact is disposed in a backside interlayer dielectric layer.
12. The semiconductor structure according to claim 11, wherein another portion of the backside

contact is disposed on the first unmerged source/drain region.

13. The semiconductor structure according to claim 10, further comprising: gate structures disposed adjacent the first unmerged source/drain region and the second unmerged source/drain region.

14. The semiconductor structure according to claim 13, wherein each of the gate structures is part of a stack of nanosheet channel layers.

15. The semiconductor structure according to claim 14, wherein the first unmerged source/drain region is disposed on opposing sidewalls of respective nanosheet channel layers, and the second unmerged source/drain region is disposed on opposing sidewalls of respective nanosheet channel layers.

16. The semiconductor structure according to claim 11, further comprising: a backside interconnect structure disposed on the wrapped-around backside contact and the backside interlayer dielectric layer.

17. The semiconductor structure according to claim 13, further comprising a frontside interconnect structure disposed over the gate structures.

18. An integrated circuit, comprising: one or more semiconductor structures, wherein at least one of the one or more semiconductor structures comprises: a backside contact; a frontside interlevel dielectric layer disposed on a top surface of the backside contact; a first unmerged source/drain region; a frontside contact; and a second unmerged source/drain region; wherein the backside contact is wrapped-around the first unmerged source/drain region; and wherein the frontside contact is wrapped-around the second unmerged source/drain region.

19. The integrated circuit according to claim 18, wherein the at least one of the one or more semiconductor structures further comprises: a plurality of gate structures disposed adjacent the first unmerged source/drain region and the second unmerged source/drain region; wherein each of the gate structures is part of a stack of nanosheet channel layers.

20. The integrated circuit according to claim 19, wherein the first unmerged source/drain region is disposed on opposing sidewalls of respective nanosheet channel layers, and the second unmerged source/drain region is disposed on opposing sidewalls of respective nanosheet channel layers.
